BRAC UNIVERSITY

Department of Computer Science and Engineering

CSE 461: Introduction to Robotics

Examination: Quiz I **Duration: 25min**
Semester : Summer 26 **Full Marks: 15**

Name:	ID:	Section:

1	What is the full form of USV? **Ans.**	**1**
2	Which is not a primitive of an AI agent? a) Learn b) Compile c) Sense d) Act	**1**
3	Which robot paradigm has the weakest goal driven behavior? **Ans**	**1**
4	Which paradigm suffers from local minima? a) Deliberative b) Reactive c) Hybrid	**1**
5	Which SBC is suitable when higher CPU performance is required? a) Raspberry Pi b) TinkerBoard c) Little Panda d) Jetson Nano	**1**
6	A chemical manufacturing plant that handles **hazardous, high-temperature materials** recently introduced two coordinated robots to maintain its safety tanks and pipelines. One compact robot operates inside the factory's pipe network, navigating tight spaces to detect cracks and leaks, while a second, external robot follows its reports and uses a precision robotic arm to weld and seal the damaged areas. This automated system has significantly improved safety, reduced maintenance time, and saved the company substantial money and energy. However, the efficiency of the robotic solution also led to a major downsizing of	**10**

	the maintenance workforce, raising concerns about job displacement alongside technological progress. A. **Which** transmission system is the least suitable for the compact robot and **why**? **[1+1]** B. Name any **three sensors** that the compact internal robot would require to work effectively. For each sensor, mention its purpose and the element it detects. **[2]** C. **What** type of motor is most suitable for the operation of the external robot described in the scenario? **Compare** your chosen motor with two other motor types and justify your selection. **[1+2]** D. Do you think it was ethically justified to replace a large portion of the maintenance workforce with robots in this factory? **Explain** your answer using the *thumb rules of robot usage.* **[3]**	

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Name:	ID:	Section:

1	What is the full form of UUV? **Ans.**	**1**
2	Which is a primitive of an AI agent? a) Navigate b) Compile c) Plan d) React	**1**
3	Which robot paradigm is similar to the reflex of biological systems? **Ans**	**1**
4	Which SBC is suitable when higher GPU performance is required? a) Raspberry Pi b) TinkerBoard c) Little Panda d) Jetson Nano	**1**
5	Which paradigm suffers from local minima? a) Deliberative b) Reactive c) Hybrid	**1**
6	A chemical manufacturing plant that handles hazardous, high-temperature materials recently introduced two coordinated robots to maintain its safety tanks and pipelines. One compact robot operates inside the factory's pipe network, navigating tight spaces to detect cracks and leaks, while a second, external robot follows its reports and uses a precision robotic arm to weld and seal the damaged areas. This automated system has significantly improved safety, reduced maintenance time, and saved the company substantial money and energy. However, the efficiency of the robotic solution also led to a major downsizing of	**10**

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Rubric

1-5	For each correct MCQ or Answer - 1
6	a. Correct system - 1 Correct reason - 1 b. 1 sensor with purpose - 1 2 sensor with purpose - 1.5 3 sensor with purpose - 2 c. Correct motor - 1 1 Comparison correct with proper explanation - 1 2 Comparison correct with proper explanation - 2 d. Properly mention the thumb rules needed for explanation - 2 Justify the reason - 1

Answers

1. USV - Unmanned/ Uncrewed Surface vehicle
   UUV - Unmanned/ Uncrewed Underwater vehicle

2. Primitives - Sense, Plan, Act, Learn
3. Similar to reflex - Reactive paradigm
4. Weakest goal driven - Reactive
5. Suffers from local minima - Reactive
6. CPU performance - TinkerBoard
   GPU - Jetson Nano

7. A. Belt is the least suitable in this case (environmental resistance low)
   B.

Camera	Detect Cracks
Sonar/Ultrasonic	Detect cracks, obstacles, navigation
Lidar	Mapping, cracks, navigation
Chemical sensor	Smell, chemical

Proximity	Obstacles, navigation
Thermometer	Temperature
Gyroscope/Accelerometer	Movement, Inclination, orientation
Barometer/ Pressure gauge	Pressure

C. Servo motor perfect for precision

- Comparison given in slides

D. Both are correct with proper explanation. Thumb rules must be mentioned. Rules given in slides.

-Justified considering the amount of positive aspects other than human layoff

- Not justified considering we prioritize humans first and people are willing to work.